# CLEF 2026 ImageCLEF Deepfake — 예상 질문 & 답변

Team Go To Germany · 정현 발표용 준비 노트 · 2026-09-20 · 리허설 피드백(2026-09-17) + PGD/DI/MI/EOT 개념 + 청중 예상 질문 포함

## Part 1 · 리허설에서 교수님이 하신 질문 (2026-09-17)

### Q1. FLUX+PuLID와 ControlNet을 같이 썼는데, 각각 어떤 정보를 유지하려고 사용한 건가요?

**[한국어 요지]**

PuLID는 대회 평가지표 중 신원 유사도(identity similarity)를, ControlNet은 랜드마크 거리(landmark distance)를 담당한다. 두 모듈이 대회 지표 두 개에 하나씩 대응됨.

**English answer**

The task scores two things: identity similarity and landmark distance. PuLID serves the first — it injects the target person's identity from a small set of reference face crops. ControlNet enforces the second — it constrains the output to match the input face landmark map. Both conditions are injected into the same FLUX diffusion transformer during denoising, so they act jointly rather than sequentially.

### Q2. ControlNet을 빼면 성능 차이가 크게 나나요? PuLID를 빼면?

**[한국어 요지]**

ControlNet 제거 → 랜드마크 거리 점수 붕괴(같은 사람이지만 다른 포즈·표정). PuLID 제거 → 자연스러운 얼굴이지만 다른 사람 → 신원 유사도 0에 가까움. 둘 다 필수.

**English answer**

In our early tests, removing ControlNet made the landmark distance score collapse — the generated face was the right person but in the wrong pose or expression. Removing PuLID made the face plausible but no longer identifiable as the target, which zeroed out the identity similarity score. Both components are essential.

### Q3. 두 조건이 서로 상충(trade-off)하는데 어떻게 밸런싱 했나요?

**[한국어 요지]**

PuLID weight ↑ → 신원 ↑ / 랜드마크 ↓, ControlNet strength ↑ → 반대. 변수 4개(PuLID weight, ControlNet conditioning scale, guidance scale, control-guidance end step)로 약 30가지 조합을 grid search. 최종 PuLID weight = 0.85. 추가로 identity당 11개 seed 생성 → landmark outlier(0.02) 제거 → identity 유사도 최고 선택.

**English answer**

There is a trade-off: raising PuLID weight improves identity but distorts landmarks; raising ControlNet strength does the opposite. We searched about thirty combinations across four hyperparameters — mainly PuLID weight, ControlNet conditioning scale, guidance scale, and control-guidance end step — and kept the combination that maximized the joint score. The final PuLID weight is 0.85. On top of that, we generate eleven seeds per identity, drop landmark-outlier candidates using a 0.02 threshold, and keep the one with highest identity similarity.

### Q4. 신원 주입(Identity Injection)이 언제 이뤄지나요? 디퓨전 후에? 도중에?

**[한국어 요지]**

Denoising 도중(during)에 들어감. 참조 얼굴 크롭 최대 20장 → 평균 identity embedding → FLUX transformer의 각 cross-attention layer에 매 denoising step마다 삽입. ControlNet 랜드마크 조건과 함께 동시에 작용.

English answer

Identity injection happens during denoising, not after. From about twenty reference face crops of the target, PuLID computes an averaged identity embedding. That embedding is inserted into every cross-attention layer of the FLUX diffusion transformer at each denoising step, alongside the ControlNet landmark condition. So the identity and pose constraints are enforced jointly through the entire generation, not applied to a completed image.

## Q5. 디텍터 우회는 파이프라인 어디에서 이뤄지나요? 노이즈만 주입? 최종 이미지가 다시 루프를 타는 건 없나요?

[한국어 요지]

FLUX가 clean fake 생성 완료 → 그 다음 PGD 25 iteration. 매 iteration마다: 이미지 → 12개 탐지기 동시 → fake probability 최소화 방향 gradient → α=0.5/255 step → clean image 기준 ε=2/255 안으로 projection → 반복. 이미지가 attack loop를 도는 것이지 FLUX를 다시 타는 것은 아님.

English answer

After FLUX finishes generating the clean fake image, we run a projected gradient descent attack for 25 iterations. Each iteration passes the image through twelve detectors simultaneously, gets gradients that minimize their fake probability, and updates the image by one step of size 0.5 over 255, clipped to stay within 2 over 255 of the original. The image itself loops through the attack, not through FLUX again. The output is the clean fake plus an imperceptible perturbation.

## Q6. Decision rule에서 threshold 0.5와 0.04는 왜 그렇게 정했나요?

[한국어 요지]

한 detector는 AI-gen, 다른 하나는 face-manipulation 담당. max(p1, p2) > 0.5는 표준 threshold — 둘 중 하나라도 확신하면 fake. min(p1, p2) > 0.04는 둘 다 약하게라도 fake라고 보면 잡음. 0.04는 held-out 7,000장 보정 세트에서 count-based net gain +433을 준 값.

English answer

One detector covers AI-generated images, the other face manipulations. So we call an image fake if either is confident — max probability above 0.5 — or if both lean fake at once — min probability above 0.04. We calibrated 0.04 on a held-out set of seven thousand images where it gave the largest count-based net gain of plus 433. On the official test set it did not transfer as well, which we discuss on the next slide.

## Q7. 재보정(recalibration) 후 성능은 어떤가요?

[한국어 요지]

공식 test set GT가 공개되지 않아 직접 재계산 불가. 다만 internal validation에서 AI-gen specialist AUROC 0.995, face-manip specialist 0.810 이었으므로, 제대로 된 rate-based calibration이면 real-side accuracy는 90% 이상 가능했을 것. 병목은 threshold 선택이었지 classifier 자체가 아님.

English answer

The ground truth for the official test set has not been released, so we cannot directly recompute the score with a new threshold. What we can say is that our AI-generation specialist has AUROC 0.995 and our face-manipulation specialist 0.810 on internal validation, meaning proper rate-based calibration should lift real-side accuracy above ninety percent. The bottleneck was the threshold choice, not the underlying classifiers.

## Q8. Further Experiments를 왜 하게 됐나요?

[한국어 요지]

우리 attack이 자체 detector를 뚫었음 → 자연스러운 follow-up: "입력이 공격당했는지 우리가 알 수 있을까?" 대회 요구 사항은 아니었지만 논문 리뷰어들이 이 확장을 좋게 봄. 세 가지 시도 → 앞 둘은 실패, 셋째(purification)가 성공.

**English answer**

Our own attack broke the detectors, which raised a natural follow-up question: can we tell when an input has been attacked? The competition did not ask this, but we thought defenders would need it. We tried three approaches — multi-detector instability, Mahalanobis distance, and input-space purification. The first two failed in deployment, but purification worked: the absolute delta-logit under median-3 filtering, through the EFFORT detector, separates attacked from clean images with AUROC 0.81 to 0.98 across four source types.

# Part 2 · 기술 개념 심화 질문

## Q9. PGD가 뭔가요? BIM(Basic Iterative Method)이랑 뭐가 다른가요?

**[한국어 요지]**

공통점: 둘 다 반복 gradient sign attack. 매 스텝 loss(=fake probability 감소 방향) gradient sign으로 조금씩 노이즈 주입.

핵심 차이: PGD는 clean image + 작은 random uniform noise로 초기화, BIM은 clean image 그대로에서 시작.

왜 중요한가: Random start가 있으면 ε-ball 내 여러 지역해로 수렴 가능. "clean point 근방 gradient만 flatten"하는 방어에 강함. 결과적으로 attack success rate가 더 높음.

경계 제약: 둘 다 매 스텝 후 clean image로부터 L∞ 거리 ε 안으로 projection. 그래서 "Projected" GD.

**English answer**

PGD and BIM are both iterative gradient sign attacks that step the image toward maximizing the detector's loss, then project it back into an epsilon-ball around the clean image. The main difference is initialization: PGD starts from the clean image plus a small random uniform noise, while BIM starts from the clean image itself. Random initialization makes PGD explore more of the epsilon-ball and gives higher attack success rates, especially against defenses that only flatten gradients near the clean point.

## Q10. DI (Input Diversity)가 뭔가요?

**[한국어 요지]**

매 PGD step마다 확률 p=0.5로 이미지를 랜덤 리사이즈 & padding한 후 그 위에서 gradient 계산.

Perturbation이 정확한 입력이 아니라 이미지 변환의 분포에 대해 최적화됨.

우리 세팅에서는 12개 white-box detector의 세부 지형에 overfit하는 것을 막고, held-out participant model 5개에 대한 전이성(transferability)을 확보. 이게 held-out 58% evasion의 핵심 요인 중 하나.

**English answer**

Input Diversity randomly resizes and pads the image with probability 0.5 at each PGD step before computing gradients. The perturbation is optimized against a distribution of image transformations rather than the exact input. This prevents the attack from overfitting to one detector's specific geometry and improves transferability to unseen participant detectors — that's what enables our 58 percent evasion on held-out models. Without DI, transfer would be closer to chance.

## Q11. MI (Momentum Iterative)가 뭔가요?

**[한국어 요지]**

Vanilla PGD는 매 step gradient를 즉시 사용 → 특정 detector loss surface의 좁은 골에 overfit되기 쉬움.

MI는 gradient를 momentum μ=1.0으로 누적 후 그 방향으로 업데이트. SGD → SGD with momentum과 같은 관계.

결과: gradient trajectory 안정화, 여러 detector에 공통으로 좋은 방향(fake→real)이 강조됨, 좁은 지역해 탈출.

**English answer**

Momentum Iterative accumulates gradients across iterations with a momentum coefficient — we use mu equals 1.0 — instead of using each step's raw gradient directly. This is analogous to SGD with momentum. It stabilizes the update direction, escapes narrow local minima specific to one detector, and empirically increases both attack success and transferability. Combined with input diversity, it acts as an implicit expectation over transformations.

## Q12. EOT (Expectation over Transformation)가 뭔가요? Explicit이 아니라 Implicit이라는 게 무슨 뜻인가요?

**[한국어 요지]**

Explicit EOT: 매 스텝에서 여러 transformation T를 sampling하고 E_T[∇f(T(x))]을 계산. Transformation 분포에 강건한 perturbation을 학습. 단점: 스텝당 여러 forward pass가 필요 → 12 detector × K transformation → 매우 비쌈.

Implicit EOT (우리 방식): DI가 스텝마다 transformation을 sampling하는 역할, MI가 여러 step에 걸쳐 gradient를 averaging하는 역할. 두 기법을 조합하면 스텝당 forward 1번으로 explicit EOT와 유사한 효과.

**English answer**

Explicit EOT computes the expected gradient over a distribution of transformations at each step, requiring multiple forward passes per iteration. That's expensive with twelve detectors in the loop. We approximate it implicitly: input diversity samples a random transformation per step, and momentum averages gradients across iterations. Together they yield the same effect — perturbations that are robust across an implicit distribution of resizes, paddings, and JPEG compressions — at the cost of a single forward pass per iteration.

## Q13. DiffJPEG은 뭔가요? 왜 in-loop로 넣었나요?

**[한국어 요지]**

실제 detector 배포는 대부분 JPEG-compressed 이미지에서 이뤄짐. Perturbation이 JPEG 압축 후에도 살아있어야 함.

표준 JPEG는 quantization step에서 미분 불가능 → gradient가 흐르지 않음.

DiffJPEG은 quantization을 differentiable soft approximation으로 대체. Q80 in-loop로 넣으면 attack이 "JPEG-Q80 압축 후에도 살아남는 perturbation"을 학습.

**English answer**

JPEG compression is not differentiable at the quantization step, so gradients cannot flow through standard JPEG. DiffJPEG approximates quantization with a differentiable soft function. We apply DiffJPEG at quality 80 inside every PGD iteration, so gradients penalize perturbations that get destroyed by JPEG compression. The result is that our adversarial images retain their attack strength after being saved as JPEG — which matters because real-world detector pipelines almost always JPEG-compress inputs.

## Q14. Mahalanobis에서 five rescue attempts는 각각 뭐였고 왜 다 실패했나요?

**[한국어 요지]**

Mahalanobis distance = 훈련 데이터의 클래스별 feature 분포를 다변량 Gaussian으로 fit하고, 새 샘플이 그 분포에서 얼마나 벗어났는지 측정. 원 아이디어: "adversarial noise가 classifier head에 대해 최적화됐으므로 feature 분포의 low-likelihood 영역에 떨어질 것". Pilot AUROC = 1.0, 배포에서 distance 분포가 6.75× 커지는 shift 발생. 다섯 rescue 모두 recall ≤ 4.4%:

① Gaussian refit on test features — test로 Gaussian을 다시 fit. 테스트에는 라벨이 없어 클래스 조건부 분포 fit 불가. 전체 covariance만 fit하면 클래스 분리력 상실.

② Per-dimension z-score normalization — 각 feature 차원별로 z-score 정규화해 scale shift 흡수 시도. Shift가 dimension-wise가 아니라 전체적인 spread 확대라 흡수 안 됨.

③ Raw-distance thresholding — 절대 거리로 그대로 threshold. Shifted 분포에서는 모든 샘플이 커진 distance를 가지므로 threshold 무의미.

④ Inter-class distance ratios — "가장 가까운 클래스 / 두 번째" 같은 상대 지표. Adversarial은 특정 클래스에만 가까운 게 아니라 모든 클래스에서 멀리 있어 ratio가 정보 X.

⑤ Inverted-distance classification — "가장 먼 클래스"로 라벨링. Shifted 분포에서는 클래스별 상대 거리 자체가 정보를 잃음.

근본 원인: Mahalanobis는 fixed multivariate Gaussian을 가정. Feature 분포 자체가 shift하면 unlabeled test에서 post-hoc rescaling으로 class separation을 복구할 수 없음.

**English answer**

The pilot AUROC was 1.0, so we thought this would work. But at deployment, the feature distance distribution shifted about 6.75 times larger. We tried five rescues — refitting the Gaussian on test features, per-dimension z-score normalization, raw-distance thresholding, inter-class distance ratios, and inverted-distance classification. All yielded 4.4 percent recall or lower. The root cause is that Mahalanobis assumes a fixed multivariate Gaussian, and when the underlying feature distribution itself shifts, no post-hoc rescaling recovers the class separation without labeled test data.

# Part 3 · 청중이 슬라이드 보고 할 법한 예상 질문

## § Generation

### Q15. Why 25 iterations for PGD? Why not more?

We saw diminishing returns beyond twenty iterations on our internal twelve white-box detectors — attack success plateaued around 95 percent. Beyond that, extra iterations mainly increase overfitting to the ensemble rather than improve transfer. Twenty-five is a compromise between white-box saturation and generation time — the whole pipeline runs at about thirty seconds per image.

### Q16. Why epsilon 2 over 255? Isn't that very small?

The task evaluates on perceptual quality as one of its metrics, and epsilon 2 over 255 is well below the visibility threshold for typical viewing conditions — that's why the perturbation looks essentially black at unit gain in the pipeline figure. Larger epsilon would improve attack success but hurt our image-quality score. Our 0.6188 image-quality result confirms perturbations were essentially invisible.

### Q17. How did you pick the twelve white-box detectors?

We took every publicly available deepfake detector we could load — the well-cited backbones like SigLIP, DINOv2, CLIP-based binary classifiers, and DeiT variants. Then we removed any that gave near-random performance on our clean FLUX outputs, since attacking a detector that already can't tell the difference gives no useful gradient signal. The final twelve all reached above 0.7 fake probability on clean fakes.

### Q18. What was the transferability gap and why so big?

White-box evasion was above 95 percent on fifteen of seventeen detectors, but held-out participant models sat at 58 percent. Gradient-based attacks fundamentally optimize against the loss surfaces they can see. Held-out models differ in backbone family, training data, or fine-tuning objective. Input diversity, momentum, and DiffJPEG were chosen specifically to close this gap — without them transfer would be near chance.

### Q19. Why FLUX over Stable Diffusion XL?

FLUX.1-dev has stronger prompt adherence and identity preservation out of the box, and the PuLID and ControlNet-Union-Pro-2.0 ecosystems for FLUX are mature — so we could combine identity and pose control without training. SDXL would have required either fine-tuning or additional custom adapters.

### Q20. Have you tried adaptive attacks against your own detection pipeline?

For internal validation, yes — our twelve white-box detectors were what the attack optimized against, so those results are already adaptive. For the further-experiments purification signal, we have not run an adaptive attack that specifically tries to survive median-3 filtering. That's an obvious follow-up. A defender-aware attacker could design lower-frequency perturbations that are median-invariant.

## § Detection

### Q21. Why not train a single unified detector instead of an ensemble?

We tried single detectors and each had a clear failure mode — AI-generation specialists missed face-manipulation artifacts, and vice versa. Rather than train a new joint model on limited data, we combined two proven specialists at inference time. Faster to deploy and gives us interpretable per-branch scores.

### Q22. Why max-probability ensemble instead of averaging or learned fusion?

Averaging assumes both detectors are equally reliable on every input, which they are not — a face-manipulation image usually has a strong GenD signal and a weak SigLIP signal. Averaging dilutes that. Max preserves the informative branch. Learned fusion would need calibrated labels for both fake types on the same images, which the training data does not provide cleanly.

### Q23. Why the perceptual-hash short-circuit before the detectors?

We generated the fake images ourselves during the generation track. If an input matches one of our own submissions exactly by pHash, we know it's fake without running detectors. This trivially catches the case where organizers include our own submissions in the test set. Cost is one hash lookup per image.

### Q24. Real-side accuracy 43 to 57 percent is quite low. Why?

That's the calibration mistake we highlighted. Our threshold choice — min above 0.04 — was calibrated on a held-out set with a specific fake-to-real ratio. The official test set had a different distribution, so the threshold's false-positive rate cost us more than its true-positive gain helped. With proper distribution matching, the same classifiers would score above 90 percent.

## § Further Experiments (§6)

### Q25. Why median-3 specifically? Why not median-5 or Gaussian?

Median-3 is the smallest local operator that destroys single-pixel high-frequency noise while preserving edges. Adversarial perturbations under epsilon 2 over 255 are exactly this — pixel-level high-frequency noise. Median-5 removes more signal, which strengthens the delta-logit magnitude but also removes more content, so clean images shift too and separation degrades. Gaussian blurs across the frequency spectrum and doesn't target the specific noise structure.

### Q26. What exactly is the JPEG confound?

JPEG compression also removes high-frequency content, especially at lower quality. So a clean JPEG-compressed image, passed through median-3, shows a smaller delta-logit than an uncompressed clean image — but still nonzero. Adversarial images and low-quality JPEG images both produce large delta-logit under median-3, making them indistinguishable. Our single-filter separation is 0.99 for clean-vs-adversarial but only 0.83 for JPEG-Q80-vs-adversarial. That gap is the confound.

### Q27. Why does EFFORT stand out among the six CLIP fine-tuning methods?

EFFORT trains a low-rank residual on top of frozen CLIP weights using an SVD decomposition. This preserves the original feature manifold while learning a small task-specific correction. When you perturb the input, the frozen path stays mostly stable but the residual reacts strongly — giving a clean, isolable delta-logit signal. Full fine-tuning and standard LoRA change too much of the backbone, so the delta-logit becomes noisy and less separable.

### Q28. Is EFFORT + median-3 a deployable defense?

Not directly as a deployed defense — it's a diagnostic signal. On uncompressed input it separates adversarial from clean at AUROC 0.81 to 0.98 depending on source type. Below JPEG quality 70 the signal disappears. Deployment would require either upstream JPEG-quality detection or accepting that low-quality JPEG inputs cannot be flagged this way. And an adaptive attacker aware of the signal can design median-invariant perturbations.

### Q29. Is the JPEG-quality-70 cliff documented elsewhere?

We haven't seen it published under that name. What is documented is that JPEG destroys high-frequency content and interacts with any high-frequency-sensitive filter. Our contribution here is empirical — the specific cliff at Q70 and the behavior of EFFORT in that regime. This felt like a natural finding to report but we don't claim it as a new theoretical result.

### Q30. Why did Mahalanobis fail while purification worked?

Mahalanobis works in feature space with an assumed fixed Gaussian, so any distribution shift between pilot and deployment breaks it. Purification works in input space with a fixed operator — the median filter itself doesn't shift. As long as the operator's physical effect on high-frequency noise remains consistent, the signal transfers.

## § Meta / Operational

### Q31. What was the total compute for the generation phase?

About thirty seconds per adversarial image on a single A100. We generated the full submission set, so roughly 180 GPU-hours end-to-end. The PGD loop dominates — twenty-five iterations times twelve detector forward passes plus one FLUX generation.

### Q32. Will you release code and models?

Yes, we plan to release the generation pipeline and the further-experiments purification scripts after publication. All detector weights we used are already public — we cite each in the paper.

### Q33. Are you continuing this line of work?

Yes, in two directions. First, adaptive attacks against the median-3 defense to characterize its true robustness. Second, extending purification beyond spatial filters to frequency-domain and learned defenses.

# Part 4 · 어려운/공격적인 질문에 담담하게 답하기

교수님 경험담: 동양인이라고 강압적으로 질문하는 사람이 있을 수 있음. 감정적 대응 X, 웃으면서 담담하게, 필요하면 offline 처리.

## 전략 4단계

① 질문의 요지를 먼저 확인: "So you're asking whether ...?" — 잘못 이해한 채 대답하는 것 방지.

② 아는 만큼만 답, 모르면 인정: 지어내면 후속 질문에서 무너짐. 인정이 더 프로답게 보임.

③ 부분 답 + 후속 약속: "In our experiments we observed X. Whether it generalizes broadly is an open question."

④ Offline으로 연결: "I'd rather give you a proper answer by email than guess now. Could you share your email after the session?"

## 바로 쓸 수 있는 문장 템플릿

"That's a fair point — we didn't test that specifically. In our setup we found ..., but I can't confirm whether it holds more generally."

"You're right that this is a limitation. We flag it in the paper as future work."

"I don't have a number for that on hand, but I'd be happy to follow up by email if you leave your address."

"That's outside what our experiments covered — an interesting direction for the follow-up work we're planning."

"Could you clarify the question? I want to make sure I answer what you're actually asking."

"Let me answer the technical part now, and I'll send you the exact numbers afterward."

## 자주 나올 수 있는 공격적 프레임 & 대응

### QA. "Your evasion rate on held-out models is only 58 percent — that's not a very strong attack."

"You're right that white-box-to-black-box transfer is the hardest part. 58 percent on completely unseen architectures is above chance and above the typical baseline in the transfer-attack literature, but I agree it's the main limitation. Closing that gap without white-box access is an open research question, and we discuss it in the paper."

### QB. "Your detection score 0.6986 is worse than random on the real-side. Why should we take your ensemble seriously?"

"The classifier itself is strong — AUROC 0.995 on AI-generated content and 0.810 on face manipulations. Our final score was hurt by threshold calibration on a held-out set with a different distribution than the official test set. That's a calibration mistake, not a classifier failure. With rate-based calibration the same models score above 90 percent."

### QC. "You're just combining off-the-shelf modules. What's novel here?"

"The generation pipeline itself is a strong combination of off-the-shelf tools, and we don't claim novelty on that. The novelty is on two other axes: first, the custom twelve-detector PGD attack with DiffJPEG-in-loop and implicit EOT that transfers to unseen participant models. Second, the further-experiments finding that a simple weight-free signal — absolute delta-logit under median-3 through EFFORT — separates adversarial from clean at AUROC 0.81 to 0.98 across four source types, which we haven't seen reported."

### QD. "Median filter is trivial — of course it removes high-frequency noise. This isn't a real defense."

"You're right that the operator is simple. Our contribution isn't the operator — it's the systematic characterization: which detector family isolates the signal cleanly, at what JPEG-quality the signal survives, and how it compares across four adversarial source types. Simple operators combined with the right detector give a portable, weight-free diagnostic — that's the finding."